

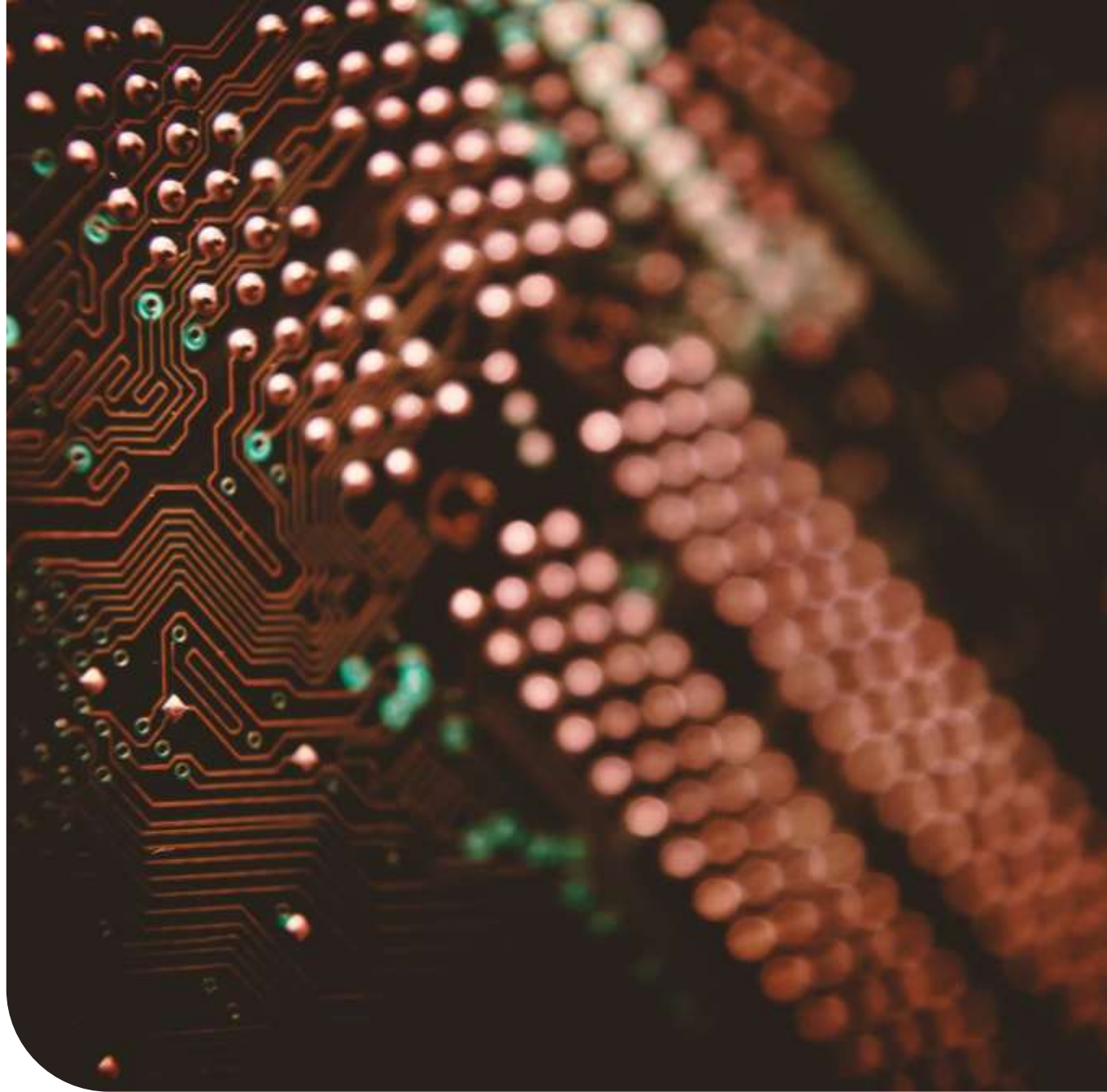


THE ROBOT PROBLEM

- You're building a **robot monitoring system**.
- Sensors keep sending values every second.
- Can you predict how many values will come?
- Yes → Array
- No → ?

THE REAL-WORLD

- Real systems deal with **unknown data sizes**
- Static structures often fail
- Industry prefers **flexible data structures**



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ARRAY LIST

QUICK RECAP — ARRAY

- Fixed size
- Same data type
- Fast access
- Example: Fixed number of motors in a robot



ARRAY LIMITATION (THINK)

What if more data arrives?

```
int[] data = new int[3];  
data[0] = 10;  
data[1] = 20;  
data[2] = 30;
```

```
// data[3] = 40; ❌ ERROR
```

SOLUTION: THE ARRAY LIST

Dynamic size

Automatically resizes

Part of Java Collections

Key Idea: "ArrayList = Smart Array"

REAL-LIFE EXAMPLE

Scenario: Student Registration

Students keep enrolling

You don't know final count

Use ArrayList

CREATING AN ARRAYLIST

```
import java.util.ArrayList;

public class CreateExample {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
    }
}
```


ADDING ELEMENTS

```
import java.util.ArrayList;

public class AddExample {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();

        list.add(10);
        list.add(20);
        list.add(30);

        System.out.println(list);
    }
}
```

1. Automatically grows
2. Real-life: Adding new users to an app



ACCESSING ELEMENTS

```
import java.util.ArrayList;

public class AccessExample {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();

        list.add(100);
        list.add(200);

        System.out.println("First element: " + list.get(0));
    }
}
```

UPDATING ELEMENTS



```
import java.util.ArrayList;

public class UpdateExample {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();

        list.add(5);
        list.add(10);
        System.out.println(list);

        list.set(1, 50); // update index 1
        System.out.println(list);
    }
}
```



REMOVING ELEMENT

```
import java.util.ArrayList;

public class RemoveExample {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();

        list.add(1);
        list.add(2);
        list.add(3);
        System.out.println(list);
        list.remove(1); // removes element at index 1

        System.out.println(list);
    }
}
```



TRAVERSING ARRAYLIST

```
import java.util.ArrayList;

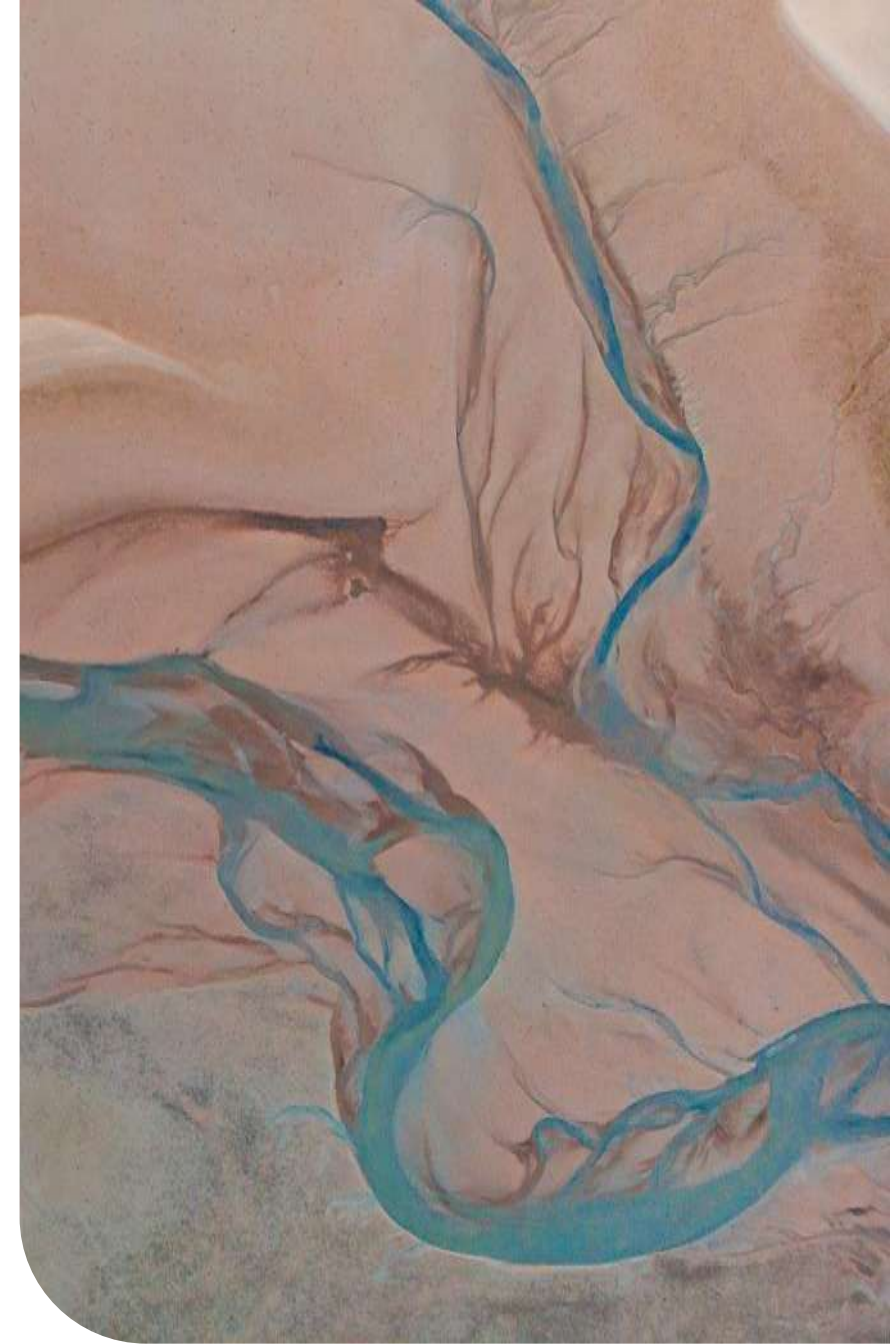
public class LoopExample {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();

        list.add(10);
        list.add(20);
        list.add(30);

        for (int i = 0; i < list.size(); i++) {
            System.out.println(list.get(i));
        }
    }
}
```


MAJOR FUNCTIONS

- `add(E e)` → adds an element at the end
- `add(int index, E e)` → inserts element at a specific position
- `get(int index)` → retrieves element at given index
- `set(int index, E e)` → updates/replaces element at index
- `remove(int index)` → removes element at given index
- `remove(Object o)` → removes the first occurrence of a value
- `size()` → returns total number of elements
- `isEmpty()` → checks if the list is empty
- `contains(Object o)` → checks if element exists in list
- `indexOf(Object o)` → returns first index of element
- `lastIndexOf(Object o)` → returns last index of element
- `clear()` → removes all elements from list
- `toArray()` → converts `ArrayList` to normal array



REAL-LIFE EXAMPLE

01: ROBOT SENSOR DATA

```
import java.util.ArrayList;

public class RobotData {
    public static void main(String[] args) {

        ArrayList<Double> sensorData = new ArrayList<>();

        // Add readings
        sensorData.add(23.5);
        sensorData.add(24.1);
        sensorData.add(22.8);

        // Print all readings
        System.out.println("Sensor Data:");
        for (int i = 0; i < sensorData.size(); i++) {
            System.out.println(sensorData.get(i));
        }

        // Update a value
        sensorData.set(1, 25.0);

        System.out.println("\nAfter Update:");
        System.out.println(sensorData);

        // Remove a value
        sensorData.remove(0);

        System.out.println("\nAfter Removal:");
        System.out.println(sensorData);
    }
}
```

REAL-LIFE EXAMPLE

02: STUDENT MARKS SYSTEM

```
import java.util.ArrayList;

public class StudentMarks {
    public static void main(String[] args) {

        ArrayList<Integer> marks = new ArrayList<>();

        // Add marks
        marks.add(80);
        marks.add(90);
        marks.add(85);
        marks.add(70);

        System.out.println("Original Marks:");
        for (int i = 0; i < marks.size(); i++) {
            System.out.println("Student " + (i + 1) + ": " + marks.get(i));
        }

        // Update a mark (correction)
        marks.set(3, 75);

        System.out.println("\nAfter Correction:");
        for (int i = 0; i < marks.size(); i++) {
            System.out.println("Student " + (i + 1) + ": " + marks.get(i));
        }

        // Calculate average marks
        int sum = 0;
        for (int i = 0; i < marks.size(); i++) {
            sum += marks.get(i);
        }

        double average = (double) sum / marks.size();
        System.out.println("\nAverage Marks: " + average);
    }
}
```


REAL-LIFE EXAMPLE

03: SHOPPING CART SYSTEM

```
import java.util.ArrayList;

public class ShoppingCart {
    public static void main(String[] args) {

        ArrayList<String> cart = new ArrayList<>();

        // Add items
        cart.add("Milk");
        cart.add("Bread");
        cart.add("Eggs");
        cart.add("Rice");

        System.out.println("🛒 Initial Cart:");
        printCart(cart);

        // Remove an item (user removes bread)
        cart.remove("Bread");

        System.out.println("\n🛒 After Removing Bread:");
        printCart(cart);

        // Add new item
        cart.add("Butter");

        System.out.println("\n🛒 After Adding Butter:");
        printCart(cart);

        // Check if item exists
        if (cart.contains("Milk")) {
            System.out.println("\nMilk is in the cart.");
        }
    }

    // Method to display cart items
    public static void printCart(ArrayList<String> cart) {
        for (int i = 0; i < cart.size(); i++) {
            System.out.println("- " + cart.get(i));
        }
    }
}
```